

LAMPNODE · DESIGN DECISIONS REGISTER

LampNode design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lampnode		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions, items to confirm, value engineering and decisions made; budget treated as a value-engineering target

LampNode design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Open decisions, Proposed, awaiting Amish.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes P1 to P11	Accept as made; or ask for changes item by item	Accept	Every component of the build plan	LPN-DDR-003
2	Mains board of the first prototype	(a) hand-wired prototype board, bench work only, laid-out board before any outdoor or luminaire-powered test; (b) lay out the board first	(a)	Mains board (build plan section 3.2) and safety stops	LPN-DDR-003, A1
3	M12 socket on the dome pad instead of the base, and the renders that still show it on the base	(a) accept and update the appearance model and renders on Amish's Mac; (b) look for a bought base with a socket boss	(a)	Dome and M12 socket; photoreal renders	LPN-DDR-003, A2
4	First co-design partner (city, utility or campus) and region	Any partner Amish chooses	None made	Not part of the TRL 3 build; sets the radio band (868 or 915 MHz antenna and module) and the driver models to test	LPN-DDR-001, O1
5	Sensor port pinout of the two expansion ports	Pinout agreed with FieldNode and the adopting teams	None yet	Port lead wiring on the head board	LPN-DDR-002 cross-repo actions; FieldNode O2
6	Appearance-model differences recorded for the product renders (compact luminaire head, cable route, split band clamps, folded bracket, added details)	Accept for the renders; or redraw	Accept; items 3 and 4 now match the constructable design	Renders only	<code>docs/REVIEW.md</code> , session 2026-09-26

■ To confirm when parts are bought

Table 2. Assumed sizes and ratings to check against the real parts.

#	What to confirm	Why it matters	Source
1	The base kit has solid material or bosses at the six screw positions on the 60 mm circle, and its blades have crimp or solder terminals inside	The dome and board stack screw through it	LPN-DDR-003, P2, P7
2	The head box's four internal bosses are 94 x 54 mm apart and 6 mm long, and its lid is 15 mm deep	They set the internal plate holes and the radar height	LPN-DDR-003, P8
3	The supply module fits 23 x 38 x 18 mm and the relay 18 x 24 x 20 mm, with an inrush rating of 80 A or more	The mains board layout and the notches depend on these sizes; R3 needs the rating	LPN-DDR-003, P6; LPN-CAL-001
4	The M12 panel socket's thread suits a 6 mm wall and a 16.2 mm hole	The dome pad is 6 mm thick	LPN-DDR-003, P3
5	The radar module fits 50 x 44 mm and has two mounting holes	The cradle holes are drilled to suit	LPN-DDR-003, P8
6	The band clamp size: the band loop is about 190 mm for a 50 mm arm, 220 mm for a 60 mm arm and 280 mm for an 80 mm arm (worm-drive clamps of about 60 to 90 mm nominal range)	One clamp size may not cover 50 to 80 mm arms	LPN-DDR-003, P10
7	The flexible antenna's datasheet allows sticking it to 3 mm of plastic	Range and tuning	LPN-DDR-003, P5
8	The driver model's inrush and its open-line dimming behaviour (full output with the 0 to 10 V line open)	Fail-on behaviour (R12) and relay sizing (R3)	LPN-CAL-001; LPN-PRC-001

■ Value engineering

Value-engineering target: USD 150 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 143.00 (USD 7.00 under the target). Main cost drivers and savings worth trying:

- The largest lines are the controller board with its LoRaWAN module, clock and coil drive (USD 25), the 24 GHz radar (USD 18), the twist-lock base kit (USD 10), the M12 socket and cordset (USD 10), and the supply and head board (USD 8 each).
- Making the design constructable added line 17, the controller boards and fixings (USD 7), and line 18, the printed head plate (USD 4), and raised lines 15 and 16 by USD 1 each; the total rose from USD 130 to USD 143.
- Savings worth trying: one laid-out board carrying both the mains and controller sides would replace two prototype boards, the harness and most of line 17; a cable glanded straight into the dome would save the M12 socket and plug (about USD 6) but make the head harder to swap; radar module prices fall steeply above prototype quantities.

■ Decisions made

Table 3. Decisions made, with Amish's words where recorded.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: ANSI C136.41 socket first, 24 GHz Doppler radar, cable to a head under the arm, LoRaWAN on the STM32WL-class module, 0 to 10 V dimming, fail-on normally closed relay, sibling sensors on FieldNode solar, photocell-equivalent default, 120 to 277 V, TALQ bridge in CityTwin	Amish: "i accept all your recommendations, go with them across all repos."	LPN-DDR-001, LPN-DDR-002
2026-09-25	R7 kept at 35 % with the reference profile and its risk accepted (O2); firmware port limit of 2.5 W above 50 °C inside (O3)	Amish, same instruction	LPN-DDR-002
2026-09-25	Engineering changes O4 to O8: TCXO clock, 80 A inrush relay with zero-cross closing, 385 V surge stage without a gas discharge tube, charge-pump coil drive and two-signal day logic, I/Q radar, 5-pole M12 ports	Amish, same instruction	LPN-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	LPN-DDR-003 (Draft, open for his review)
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register; LPN-CAL-001 v0.3